Berkeley Socket API & I/O Multiplexing

Coverage: Socket System Calls Lifecycle (TCP & UDP), I/O Multiplexing with select() and poll(), fd_set Macros, Non-Blocking Sockets

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1. Berkeley Socket API

- TCP Server: socket() -> bind() -> listen() -> accept() -> read()/write() -> close().
- TCP Client: socket() -> connect() -> write()/read() -> close(). UDP: sendto() / recvfrom().

2. I/O Multiplexing with select()

- Allows a single thread to monitor multiple descriptors simultaneously using fd_set bitmasks.

■ KTU University High-Yield Exam Questions

- Q: Explain the sequence of Berkeley socket system calls and explain select() I/O multiplexing. (8 Marks)